LUIS ANTONIO CANALES GUERRERO

Specialization: Cybersecurity · Networks · Cloud Computing · Automation

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PROFESSIONAL PROFILE

Student of Information and Communication Technology Engineering (ITICS) with a solid technical background in Defensive Cybersecurity, Network Infrastructure, and Cloud Computing. I hold certifications from industry leaders (Cisco, Microsoft, Google, Government of Mexico) that validate my technical competencies.

My approach focuses on defense in depth, combining offensive and defensive security knowledge to implement robust solutions. Practical experience applying Python and Artificial Intelligence to strengthen organizational security posture.

Core competencies: Security incident management, secure network architecture, Linux/Windows systems administration, applied cryptography, cloud security (Azure), and incident response.

EDUCATION

B.Eng. in Information and Communication Technologies (ITICS)

- Specialization in network infrastructure, cybersecurity, and distributed systems
- Practical projects: secure networks, virtualization, and security scripting
- Active member of the CRITLA Robotics Club – amateur RC minizumo competitions
- Labs on TryHackMe and Hack The Box

Technical Professional Baccalaureate in Computer Science

- Technical training in programming, networking, and operating systems
- WER Robotics competitions – representing the institution
- Automation and computer security projects

TECHNICAL CERTIFICATIONS

Cisco Networking Academy

- CCNA (Cisco Certified Network Associate) – Complete
- Python Essentials 1
- Introduction to Cybersecurity

Google & Coursera

- Foundations of Cybersecurity (Google)
- Introduction to Generative AI (Google)
- Introduction to AI for Cybersecurity (Coursera / Johns Hopkins University)

Microsoft

- Eduit – Evaluasí ECM0358 Foundations of Generative AI

Government of Mexico (CONOCER)

- Intermediate Python – Labor competency standard certification

Other Certifications

- Cybersecurity Awareness – LinkedIn Learning
- Introduction to Cybersecurity – ISC²

- MAPS Disruptivo

PROJECTS & PRACTICAL EXPERIENCE

Secure LAN Network Implementation

Full design and configuration of enterprise local network infrastructure with a focus on defensive security:

- VLAN segmentation (802.1Q) for traffic isolation by department
- ACL-based access control for filtering unauthorized traffic
- DHCP and DNS services with redundancy configuration
- Cisco device hardening: service disabling, banners, and authentication
- Static and inter-VLAN routing configuration for controlled communication

Result: Segmented network that significantly reduces the attack surface compared to traditional flat networks.

Virtual Infrastructure on Microsoft Azure

Deployment of an enterprise cloud environment following the shared responsibility model:

- VM deployment with Windows Server and Linux
- Virtual network architecture: Virtual Networks, subnets, and NSGs
- Azure Active Directory, RBAC, and compliance policies
- Azure Security Center for continuous security posture monitoring
- Azure Backup, Site Recovery, and business continuity strategies
- MFA, Conditional Access, and Privileged Identity Management (PIM)

Security Automation with Python & AI

- Python scripts for parsing, normalization, and analysis of security logs
- Basic ML algorithms for anomaly detection and suspicious pattern identification
- Automated playbooks for common incident response
- Use of LLMs for vulnerability analysis and technical report generation

CRITLA Robotics Club – ITLA

- Design and programming of RC minizumo robots for amateur robotics competitions
- Development of amateur line-follower robots: circuit design, IR sensor calibration, and control logic
- Arduino (C/C++) programming for minizumo control: combat logic, sensor reading, and motor control
- Integration of sensors (ultrasonic, IR, encoders) and actuators in embedded systems

Achievements: Medals and recognition representing ITLA at competitions at ESIME Azcapotzalco.

SPECIALIZED CYBERSECURITY KNOWLEDGE

1. Defensive & Operational Cybersecurity

Incident Management (Incident Response):

- Full cycle: Preparation → Detection → Containment → Eradication → Recovery → Lessons Learned
- Chain of custody for digital forensic evidence preservation
- Documentation per NIST SP 800-61

Monitoring & Detection Systems:

- IDS/IPS: Snort and Suricata configuration; alert analysis and proactive blocking
- SIEM: Log correlation with Splunk and ELK Stack (practical labs)
- Honeypots: Deployment of decoys for attacker behavior analysis

Defense Strategies:

- Defense in Depth, Principle of Least Privilege
- Network segmentation, micro-segmentation, and Zero Trust Architecture

2. Secure Infrastructure & Networks

Protocols & Standards:

- Wi-Fi: WEP (insecure) → WPA → WPA2-AES → WPA3 (current standard)
- TCP/IP, OSI, IPv4/IPv6, subnetting, static and dynamic routing (OSPF, EIGRP)
- VPN: IPSec, SSL/TLS, WireGuard – encrypted site-to-site and client-to-site communications
- VLANs (802.1Q), inter-VLAN routing, Private VLANs, trunking

L2/L3 Security:

- DHCP Snooping, Dynamic ARP Inspection, IP Source Guard
- STP Security: Root Guard, BPDU Guard; Port Security
- Air Gapping for critical systems (ICS/SCADA)

3. Cloud Security

- IaaS / PaaS / SaaS: customer and provider responsibilities in each model
- CASB (Cloud Access Security Broker): intermediaries between users and cloud services
- Identity management: MFA, SSO, federation, PIM, Conditional Access
- Encryption in transit (TLS) and at rest (AES-256), Azure Key Vault
- Compliance and governance: regulatory frameworks, data retention, and auditing

4. Systems Administration & Endpoint Security

- Linux: Ubuntu, CentOS, Kali Linux – administration, hardening (CIS Benchmarks), shell scripting (Bash)
- Windows Server 2016/2019 and Windows 10/11 – Active Directory, GPO, PowerShell, WSUS, BitLocker
- Virtualization: VMware, VirtualBox, Hyper-V, Docker (concepts), Kubernetes (familiarity)

5. Cryptography & Authentication

Fundamentals:

- Symmetric encryption: AES-128/256, 3DES – speed and key management
- Asymmetric encryption: RSA, ECC – PKI, digital certificates, digital signatures
- Hash functions: SHA-256, SHA-3, MD5 (insecure) – integrity and verification

Credential Protection & Authentication:

- Salting + Peppering to invalidate Rainbow Tables; Bcrypt, Argon2, PBKDF2
- MFA: TOTP, FIDO2/WebAuthn, biometrics, and physical tokens
- Attacks: Credential Stuffing, Password Spraying, Brute Force – mitigations with rate limiting and lockout

6. Physical Security & Social Engineering

- Mantraps, physical access control, RFID, biometrics, and CCTV
- Prevention of Tailgating/Piggybacking in critical areas
- Attack vectors: Phishing, Spear Phishing, Vishing, Smishing, Pretexting, Baiting
- Deepfakes and adversarial AI: detection of AI-generated identity impersonation
- Awareness programs and phishing simulations – Human Firewall

TECHNICAL SKILLS

Programming Languages

Language / Technology	Level	Applications
Python	Intermediate	Automation, data analysis, security scripting, OOP
Bash / Shell	Intermediate	Linux administration, hardening, automation scripts

PowerShell	Basic–Intermediate	Windows administration, Active Directory, auditing
C / C++	Basic	Microcontrollers, educational robotics
SQL	Basic	Database management, basic queries

Networks & Telecommunications

- TCP/IP, UDP, HTTP/HTTPS, DNS, DHCP, SNMP
- SSH, SSL/TLS, IPSec, WireGuard, OSPF, EIGRP
- Cisco routers and switches – CLI, L2/L3 routing
- VLANs (802.1Q), inter-VLAN, trunking, Port Security
- Firewalls, ACLs, NAT/PAT, site-to-site and client-to-site VPN
- Tools: Cisco Packet Tracer, Wireshark, Nmap

Cybersecurity

- NIST CSF, ISO 27001 (knowledge), CIS Controls
- Metasploit (concepts), Nmap, Wireshark, Burp Suite (basic)
- Splunk, ELK Stack (labs)
- Vulnerability analysis, risk assessment
- Incident response: containment, eradication, recovery
- Zero Trust, Defense in Depth, least privilege

Cloud Computing – Microsoft Azure

- VMs, VNets, NSGs, Azure AD, RBAC
- MFA, Conditional Access, PIM, Key Vault
- Azure Security Center, Microsoft Defender for Cloud
- Azure Backup, Site Recovery, Azure Monitor
- Storage Accounts, Log Analytics, Azure CLI
- Terraform (concepts), ARM Templates

Operating Systems & Virtualization

- Linux: Ubuntu, CentOS, Kali Linux
- Windows 10/11 and Windows Server 2016/2019
- Active Directory, GPO, IIS, WSUS, BitLocker
- VMware, VirtualBox, Hyper-V
- Docker (concepts), Kubernetes (familiarity)
- Bash scripting, PowerShell

LANGUAGES

Language	Level	Notes
Spanish	Native (C2)	Mother tongue
English	A2–B1 (Beginner–Intermediate)	Technical reading, basic documentation. Actively working toward intermediate–advanced level.

ACHIEVEMENTS & RECOGNITION

- Robotics competitions: Medals at amateur RC minizumo competitions representing ITLA at ESIME Azcapotzalco

- Pre-graduation certifications: Cisco (CCNA) and Google Cybersecurity before completing the degree
- Complete CCNA: validates advanced knowledge in networking and network security
- Active participation in cybersecurity communities: TryHackMe, Hack The Box, CTFs

CONTINUOUS LEARNING & PROFESSIONAL GOALS

In Progress

- AZ-900: Azure Fundamentals (Microsoft) – actively preparing
- CompTIA Security+ – baseline international cybersecurity certification
- Azure Security Engineer Associate (AZ-500) – next step on the Azure path
- Ethical Hacking and pentesting methodologies (OWASP, PTES, OSSTMM)
- Active labs on TryHackMe, Hack The Box, and CTF platforms

Short-Term Goals (6–12 months)

- First professional experience in a SOC (Security Operations Center) or infrastructure team
- Specialization in Cloud Security and DevSecOps practices
- Contribution to open-source cybersecurity projects

Mid- and Long-Term Goals

- CISSP (Certified Information Systems Security Professional) certification
- Enterprise security architecture and risk management
- Leadership of defensive cybersecurity teams (Blue Team)